

ChargeFlow Technologies Inc.

Technical Installation & Cost Specification

Full System Diagrams, Installation Procedure, Bill of Materials, and Supplier Directory

Ottawa, Ontario, Canada

September 2026

Note on visuals: figures in this document are original engineering schematics prepared for ChargeFlow, not photographs of a built installation (no site or product photography exists yet). Costs are sourcing-based estimates, not vendor quotes — see Section 8. Diagram labels are kept in English, the standard convention for technical/engineering schematics internationally.

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1. Purpose & Scope

This document is the technical companion to the ChargeFlow Technologies Inc. R&D Project Documentation. Where that document covers vision, market, and business planning, this one covers the engineering build: system diagrams, a full component-level bill of materials, step-by-step installation procedure, and a supplier directory with pricing and country of origin, prioritizing Canadian sourcing wherever a viable supplier exists.

Scope covers two installation tiers, matching the phased roadmap in the master documentation:

- Tier A — Bench/lab proof-of-concept (100–500 W): the near-term build this funding round supports.
- Tier B — Full site installation (single drive-thru lane or equivalent parking zone, kW-scale): a forward cost model for the Phase 3 pilot, built from the same supplier and component data, scaled up.

All costs are in CAD unless noted, current as of September 2026, and are engineering estimates built from supplier list pricing and comparable commercial systems — not binding vendor quotations. Section 8 explains the estimation basis in full.

2. System Diagrams

The three figures below illustrate the physical installation, site placement, and electrical architecture. They are original schematics prepared for this document; no built prototype or installed site exists yet, so no site photography is available.

Figure 1 — Cross-Section: Embedded Wireless Charging Installation

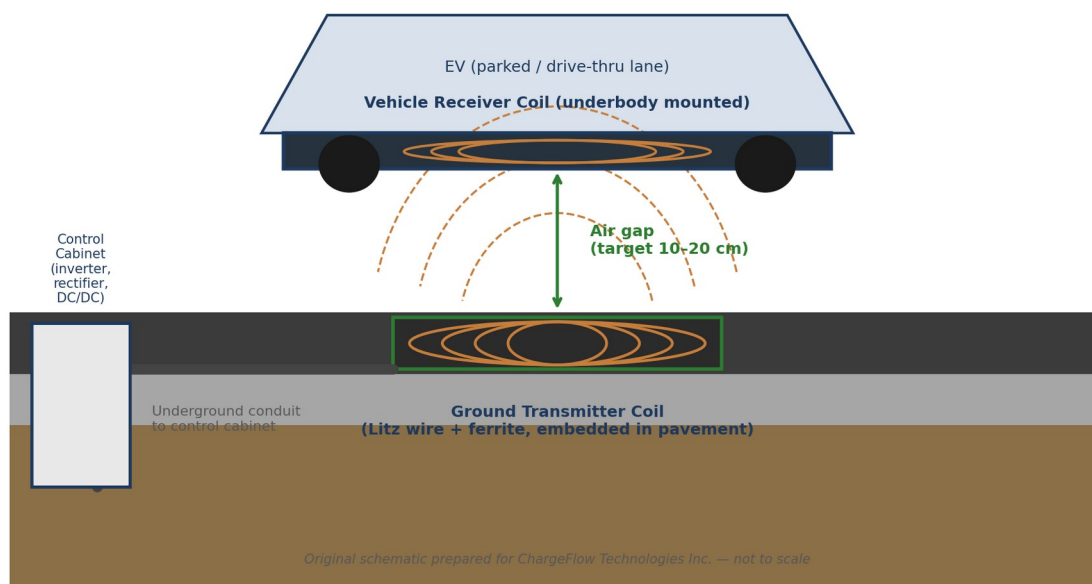


Figure 1 — Cross-Section: Embedded Wireless Charging Installation

Shows the transmitter coil embedded beneath the pavement surface, the air gap to the vehicle-mounted receiver coil, and the underground conduit run back to the control cabinet housing the power electronics.

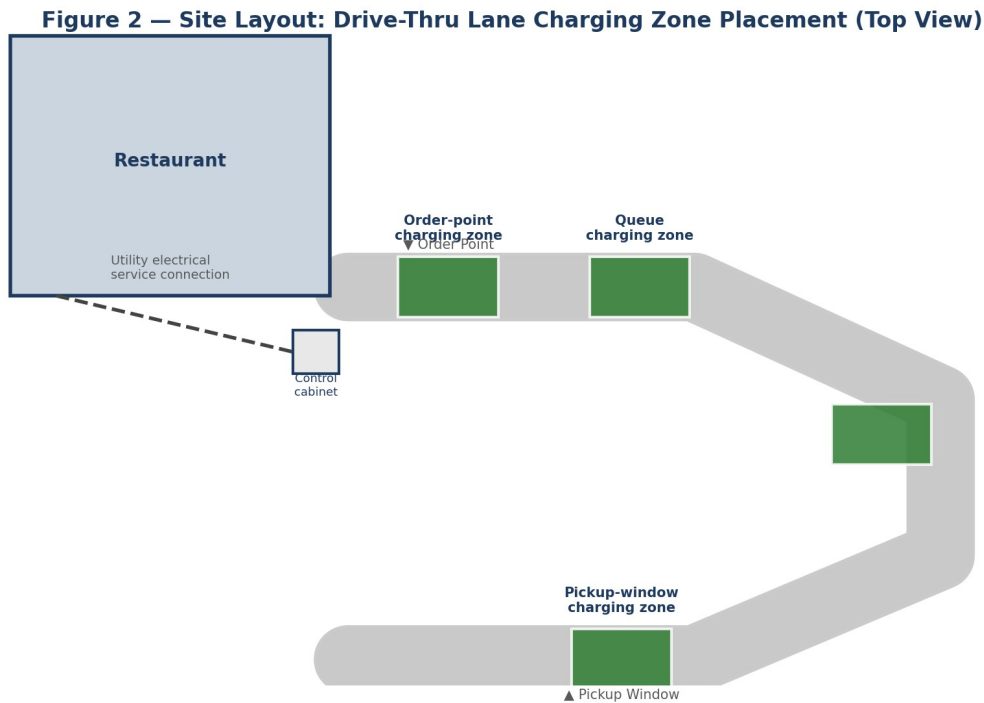


Figure 2 — Site Layout: Drive-Thru Lane Charging Zone Placement

Top-down view of a typical quick-service restaurant drive-thru lane, showing candidate charging-zone placements at the order point, queue, and pickup window, with the control cabinet positioned for a short conduit run to the building's electrical service.

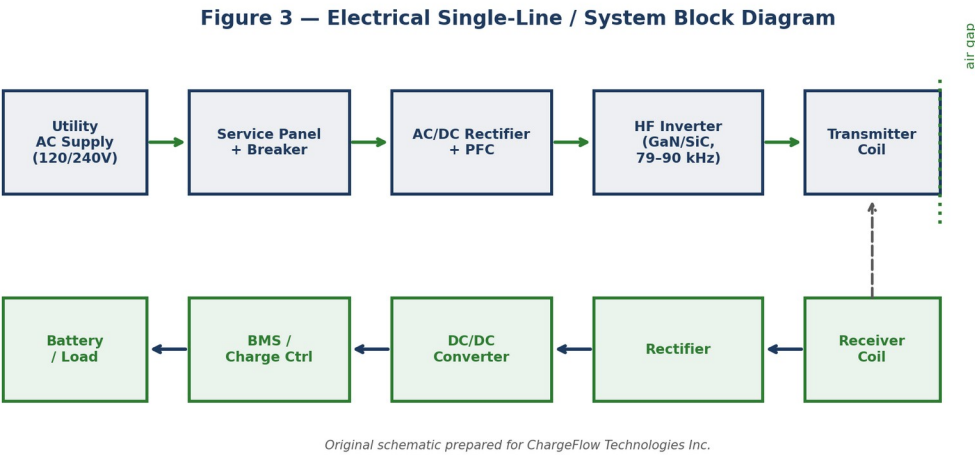


Figure 3 — Electrical Single-Line / System Block Diagram

Full power path from utility supply through to the vehicle battery, split into the ground-side chain (grid → panel → rectifier/PFC → high-frequency inverter → transmitter coil) and the vehicle-side chain (receiver coil → rectifier → DC/DC converter → battery management system → battery).

3. Component Technical Specifications

Component	Specification	Function
Transmitter coil	Litz wire winding on ferrite core plate, resonant tuned to 79–90 kHz (SAE J2954 band)	Generates the alternating magnetic field
Receiver coil	Matched Litz/ferrite assembly, vehicle-underbody rated, IP67 housing	Captures magnetic flux, induces AC voltage
High-frequency inverter	GaN or SiC half-bridge/full-bridge topology, switching at coil resonant frequency	Drives the transmitter coil
AC/DC front end	Active PFC rectifier, outdoor/enclosure-rated	Converts grid AC to regulated DC bus
Receiver-side rectifier	Synchronous or diode bridge rectifier rated to coil output	Converts induced AC back to DC
DC/DC converter	Isolated buck/boost stage, matched to target battery voltage window	Regulates output to battery-safe voltage
Control & alignment system	Microcontroller + position sensors (inductive or camera-based)	Detects vehicle position, engages/disengages transfer
Enclosure (ground-side)	NEMA 3R/4 or CSA-equivalent outdoor enclosure	Houses inverter, rectifier, control electronics
Foreign-object detection	Dedicated sensing subsystem (Phase 2+)	Detects objects/living beings between coils before energizing

4. Installation Procedure

4.1 Tier A — Bench Proof-of-Concept Setup

1. Assemble transmitter and receiver coils on a lab bench jig with adjustable vertical/lateral offset to simulate ground clearance and misalignment.
2. Wire the AC/DC front end and high-frequency inverter into the transmitter coil, confirming resonant tuning with a network analyzer or LCR meter before applying power.
3. Power up at reduced voltage first; verify waveform shape and switching behaviour on an oscilloscope before ramping to full 100–500 W target.
4. Connect the receiver coil to its rectifier and DC/DC stage, terminating into a resistive load bank or test battery.
5. Record efficiency (input W vs. output W) at the baseline aligned position, then repeat at incremental lateral and angular offsets to build the efficiency curve.
6. Verify over-current and over-temperature protection trips correctly before any unattended operation.

4.2 Tier B — Full Site Installation (Phase 3 Pilot)

7. Site electrical assessment: confirm available service capacity at the host property; engage a licensed electrical contractor (ESA-registered in Ontario) for a load study.

8. Permitting: submit electrical permit application and site plan to the municipality; schedule Electrical Safety Authority (ESA) inspection milestones.
9. Civil work: saw-cut and excavate the pavement section at the chosen charging-zone location(s); install underground conduit run from the zone to the control cabinet location.
10. Coil placement: install the transmitter coil assembly in its embedded housing at the specified depth, verify level and orientation, backfill and compact sub-base.
11. Pavement restoration: pour/patch asphalt or concrete over the embedded housing to finished grade, matching surrounding surface; allow full cure time before traffic loading.
12. Control cabinet installation: mount the CSA/NEMA-rated enclosure, install rectifier/PFC, inverter, and control electronics; land utility feed through the service panel.
13. Vehicle-side receiver installation (pilot vehicle): mount the receiver coil to the vehicle underbody, route wiring to the rectifier/DC-DC/BMS stage, integrate with the vehicle's charge port or dedicated pilot battery interface.
14. Commissioning: power up at low duty cycle, verify alignment-detection and foreign-object-detection systems, confirm safe shutdown behaviour, then run a supervised full-power charge cycle.
15. ESA final inspection and sign-off before the zone is opened to live traffic.

5. Bill of Materials & Cost — Tier A (Bench Proof-of-Concept)

Consistent with the Phase 1 budget in the master R&D documentation, itemized here with named suppliers and country of origin.

Item	Supplier (sourcing route)	Country of Origin	Est. Cost (CAD)
Litz wire (coil winding)	Patlon (Toronto, ON) — distributes New England Wire Technologies	Canada (distributor) / USA (manufacture)	\$120 – \$250
Ferrite cores/plates	Digi-Key Canada — TDK/EPCOS, Ferroxcube product lines	Canada (distributor) / Japan, Netherlands (manufacture)	\$150 – \$350
GaN power transistors (inverter)	GaN Systems (Ottawa, ON), via Digi-Key/Mouser Canada	Canada (design & engineering, Ottawa) — wafer fab international	\$100 – \$250
Rectifier diodes / SiC MOSFETs	onsemi or ROHM, via Digi-Key Canada	USA / Japan (manufacture), Canada (distribution)	\$60 – \$150
DC/DC converter module	Digi-Key/Mouser Canada — Vicor, Murata, or similar	USA / Japan (manufacture), Canada (distribution)	\$80 – \$180
Microcontroller & sensors	Digi-Key Canada / local electronics supplier	Mixed (Canada distribution)	\$100 – \$200
Bench enclosures & test rig hardware	Local Ottawa fabrication / hardware supplier	Canada	\$150 – \$300
Instrumentation access	Rental — local Ottawa test-equipment	Canada	\$150 – \$300

Item	Supplier (sourcing route)	Country of Origin	Est. Cost (CAD)
(power meter, thermal camera)	supplier or Invest Ottawa/Bayview Yards lab access		
Contingency (15%)	—	—	\$140 – \$290
Tier A Total			\$1,050 – \$2,270

This range sits within (and is more conservative than) the Phase 1 estimate in the master documentation, reflecting supplier-confirmed list pricing rather than rounded planning figures.

6. Bill of Materials & Cost — Tier B (Full Site Installation, kW-Scale Pilot)

Tier B scales the Tier A component set to kW-class hardware and adds civil, electrical, and permitting costs for a single drive-thru-lane or equivalent parking-zone installation. Hardware figures are projected from Tier A unit pricing plus benchmark data from comparable commercial systems (e.g., WiTricity Halo, an 11 kW aftermarket wireless EV charging system priced at approximately US\$3,500 per unit with US\$3,500–4,000 in professional installation, manufactured in the USA). ChargeFlow's own kW-scale hardware does not yet exist and has not been quoted by a manufacturer — this is an engineering projection, not a purchase price.

Category	Description	Supplier / Sourcing Route	Country of Origin	Est. Cost (CAD)
Charging hardware set	Transmitter + receiver coil pair, inverter, rectifier, DC/DC, scaled to several kW	Custom-built from Patlon, Digi-Key Canada, GaN Systems components (see Tier A suppliers, scaled)	Canada (assembly) / mixed imports (see Tier A)	\$4,000 – \$7,500
CSA/NEMA control cabinet enclosure	Outdoor-rated cabinet for inverter, rectifier, controls	EM Dynamics (Toronto, ON) or Hammond Manufacturing (Guelph, ON)	Canada	\$1,200 – \$2,500
Foreign-object / alignment detection system	Sensor array + control logic (Phase 2/3 scope)	Local systems integrator (Ottawa) using Canadian-distributed sensor components	Canada (integration) / mixed (components)	\$800 – \$1,800
Civil work — trenching & conduit	Excavation, underground conduit run from zone to cabinet	Local Ottawa civil/electrical contractor	Canada	\$1,500 – \$5,000
Pavement cut, patch & restoration	Asphalt/concrete removal and restoration around embedded coil housing	Local Ottawa paving contractor	Canada	\$1,000 – \$3,000
Electrical service & panel work	Service panel connection, breaker, wiring, ESA-licensed labour	Local Ottawa ESA-licensed electrical contractor	Canada	\$2,000 – \$6,000
Permitting &	Municipal electrical permit,	City of Ottawa / Electrical	Canada	\$1,000 –

Category	Description	Supplier / Sourcing Route	Country of Origin	Est. Cost (CAD)
inspection	ESA inspection fees, stamped drawings if required	Safety Authority		\$3,000
Vehicle-side receiver retrofit (pilot vehicle)	Underbody receiver installation on one pilot vehicle	In-house / local automotive electrical shop	Canada (labour) / mixed (components)	\$1,500 – \$3,000
Contingency (15%)	—	—	—	\$1,950 – \$4,770
Tier B Total (single site)				\$14,950 – \$36,570

Benchmark Comparison

For reference, a comparable commercial single-unit wireless EV charging system (WiTricity Halo, 11 kW) costs roughly \$4,800–\$5,500 CAD in hardware plus \$4,800–\$5,500 CAD in professional installation for a single home/light-commercial unit — broadly consistent with the lower half of the Tier B hardware and electrical estimate above, before ChargeFlow's added civil work for a multi-zone drive-thru lane.

7. Supplier Directory

7.1 Canadian Suppliers (Preferred Sourcing)

Supplier	Location	Supplies	Notes
Patlon	Toronto, ON	Litz wire and specialty winding solutions	Canadian distributor partnered with New England Wire Technologies (USA manufacture)
GaN Systems (Infineon)	Ottawa, ON (Kanata)	GaN power transistors for high-frequency inverters	Founded and headquartered in Ottawa; engineering/design hub remains local post-2023 Infineon acquisition
Digi-Key Canada	Distribution to Canada-wide (US-headquartered)	Ferrite cores, semiconductors, DC/DC modules, microcontrollers, sensors	Broad-line distributor; free shipping to Canada on orders over \$100 CAD; most parts manufactured abroad (Japan, USA, Germany) and distributed through a Canadian storefront
Mouser Electronics Canada	Distribution to Canada-wide (US-headquartered)	Alternate source for semiconductors, ferrites, passive components	Same sourcing profile as Digi-Key

Supplier	Location	Supplies	Notes
EM Dynamics	Toronto, ON	Custom CSA/UL-certified NEMA enclosures, including EV-charging-industry experience	100% Canadian design and fabrication
Hammond Manufacturing	Guelph, ON	Standard and custom NEMA/CSA/UL-rated enclosures	100% Canadian manufacture; over 2,000 UL/CSA-rated enclosure models
Local ESA-licensed electrical contractor	Ottawa, ON	Service panel work, wiring, permitting coordination	To be selected via local RFQ; required for any grid-connected installation in Ontario
Local paving/civil contractor	Ottawa, ON	Trenching, conduit, pavement cut and restoration	To be selected via local RFQ

7.2 International Suppliers (No Viable Canadian Source)

Component	Representative Supplier	Country	Why Sourced Internationally
Ferrite core raw material	TDK / EPCOS, Ferroxcube	Japan, Netherlands	No Canadian ferrite core manufacturer at the required specification; sourced via Canadian distributor (Digi-Key/Mouser)
Litz wire raw manufacture	New England Wire Technologies	USA	Manufactured in the USA; reaches Canada through Patlon's Toronto distribution partnership
SiC power devices (rectifier stage alternative)	onsemi, ROHM	USA, Japan	No Canadian SiC device fab; sourced via Canadian-facing distributors
DC/DC converter modules	Vicor, Murata	USA, Japan	Off-the-shelf modules with no direct Canadian manufacturer equivalent at this power class
Reference commercial benchmark hardware	WiTricity Halo	USA	Cited for cost benchmarking only — not a ChargeFlow supplier

Sourcing summary: coil winding, enclosures, civil work, electrical labour, and system integration/assembly can all be sourced within Canada — the majority by dollar value for Tier B. The main imported inputs are raw ferrite material and specific semiconductor devices, both reaching ChargeFlow through Canadian-based distributors (Digi-Key Canada, Mouser Canada, Patlon) rather than direct international purchasing.

8. Estimation Basis & Disclaimer

- Tier A costs are drawn from current (September 2026) list pricing at named distributors for components at the specified bench-scale quantities.
- Tier B hardware costs are projected by scaling Tier A component pricing to kW-class quantities and cross-checking against one publicly available commercial benchmark (WiTricity Halo); ChargeFlow has not yet built or received a manufacturing quote for kW-scale hardware.
- Civil, electrical, and permitting costs for Tier B are drawn from published Canadian EV-charging-infrastructure installation cost ranges (residential and commercial), not a site-specific quote.
- All figures should be treated as planning-grade estimates suitable for a funding application and internal budgeting — not as a fixed-price contract basis. Formal quotations from each named supplier are recommended before committing capital.

This document should be refreshed once Tier A testing produces real efficiency and thermal data, and again once a specific pilot site and contractor quotes are secured for Tier B.